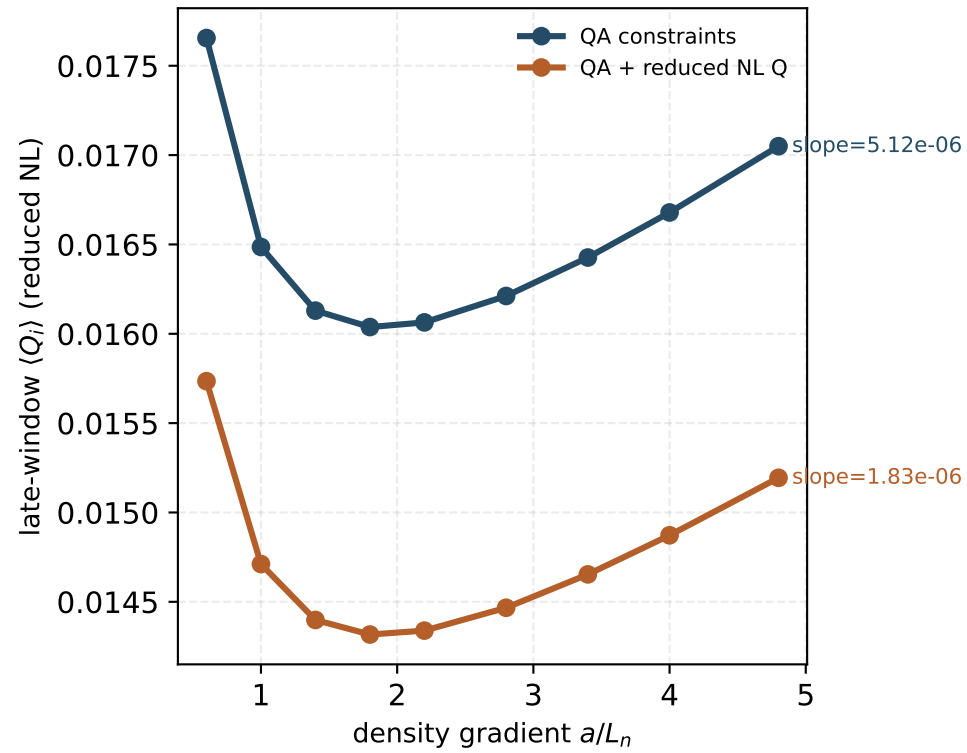
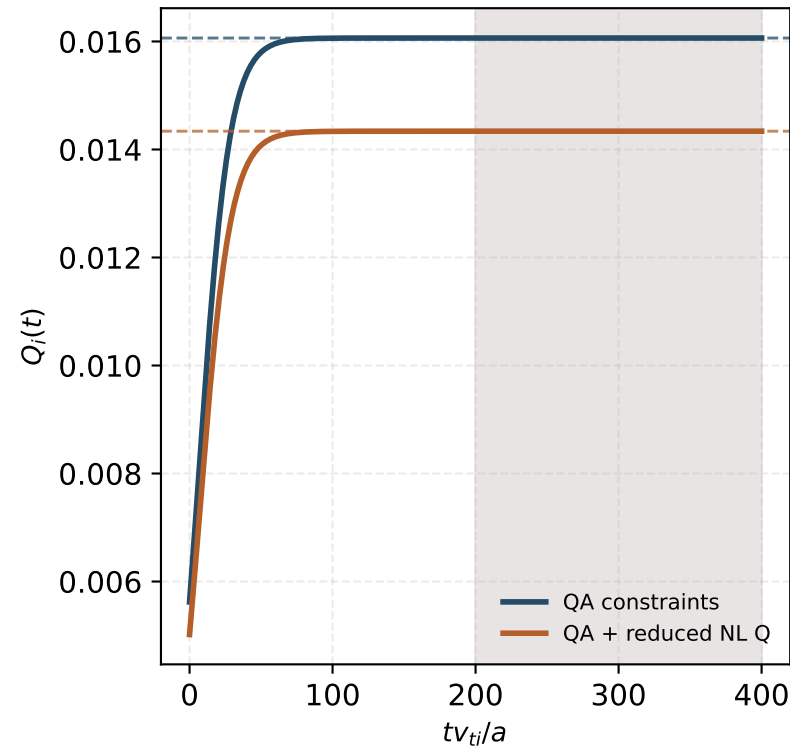


Aspect-6 QA low-turbulence optimization: constraints-only vs transport-aware design

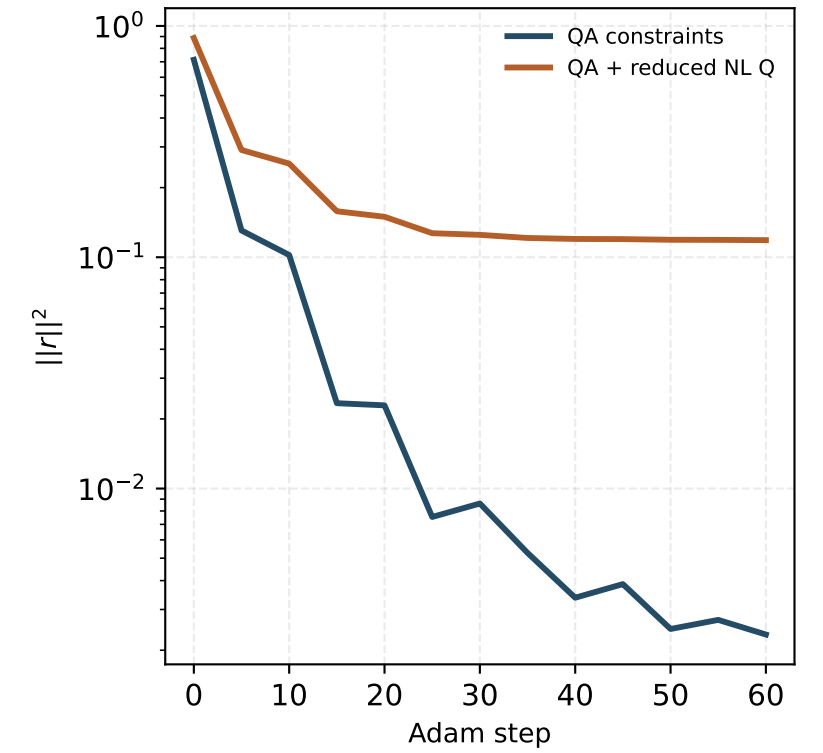
Gradient scan at fixed $a/L_{Ti} = 6$



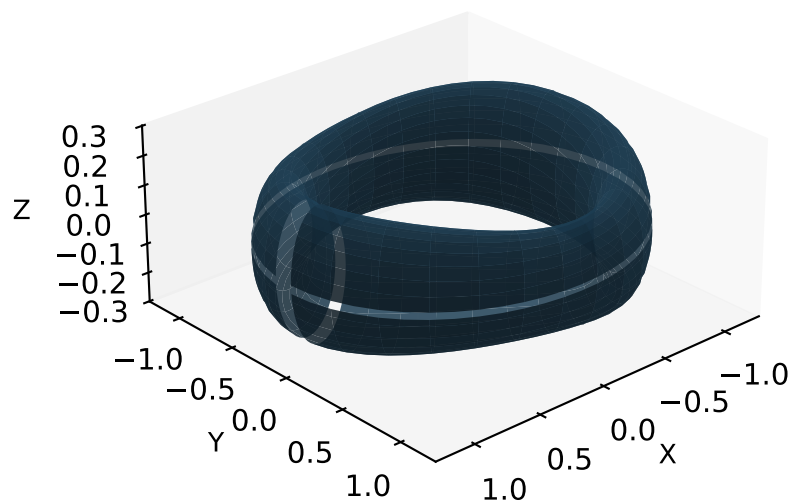
Fixed-gradient trace: $a/L_n = 2.2$, $a/L_{Ti} = 6.0$



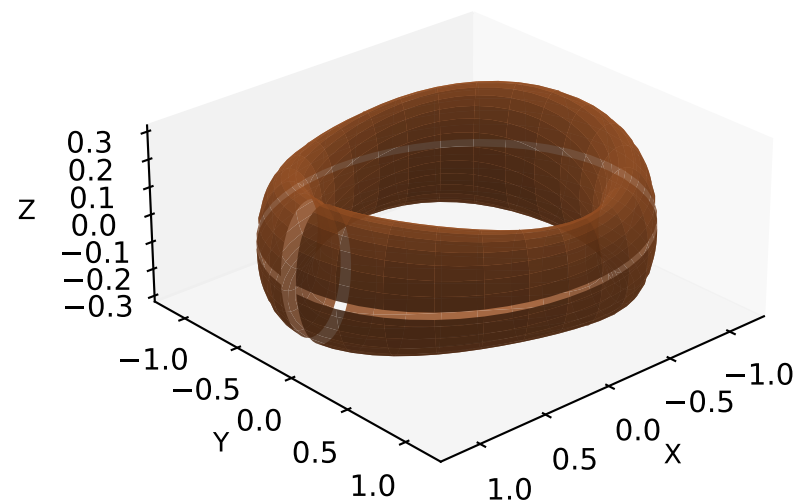
Constrained objectives



Reduced LCFS: QA constraints



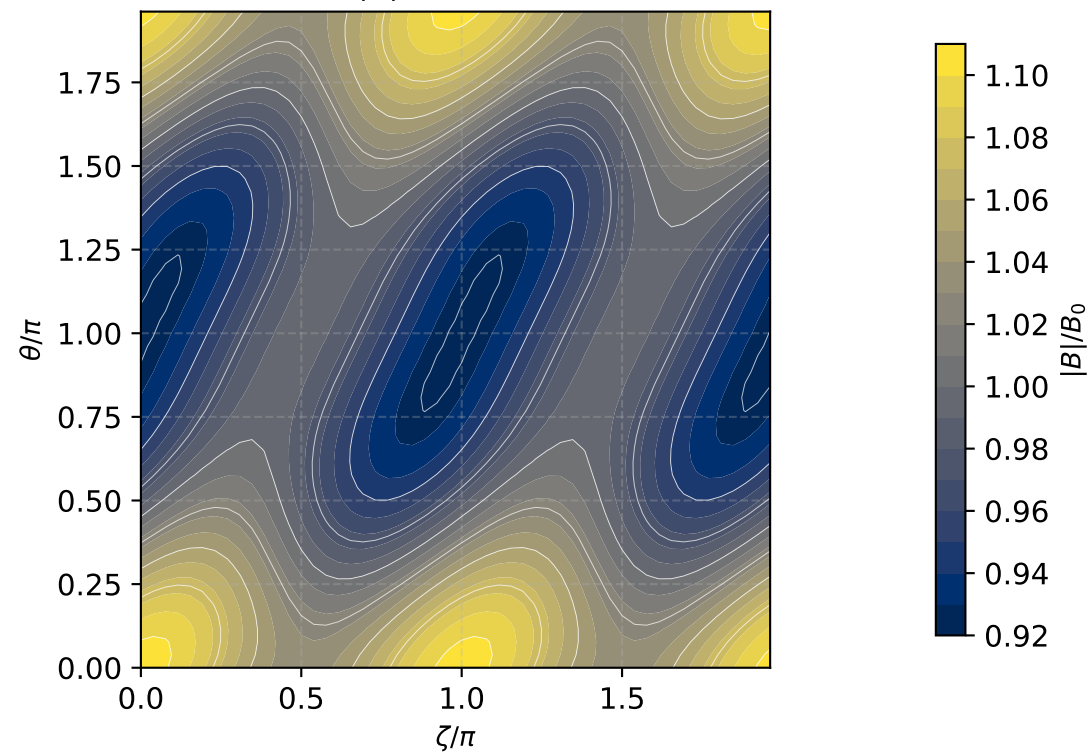
Reduced LCFS: QA + reduced NL Q



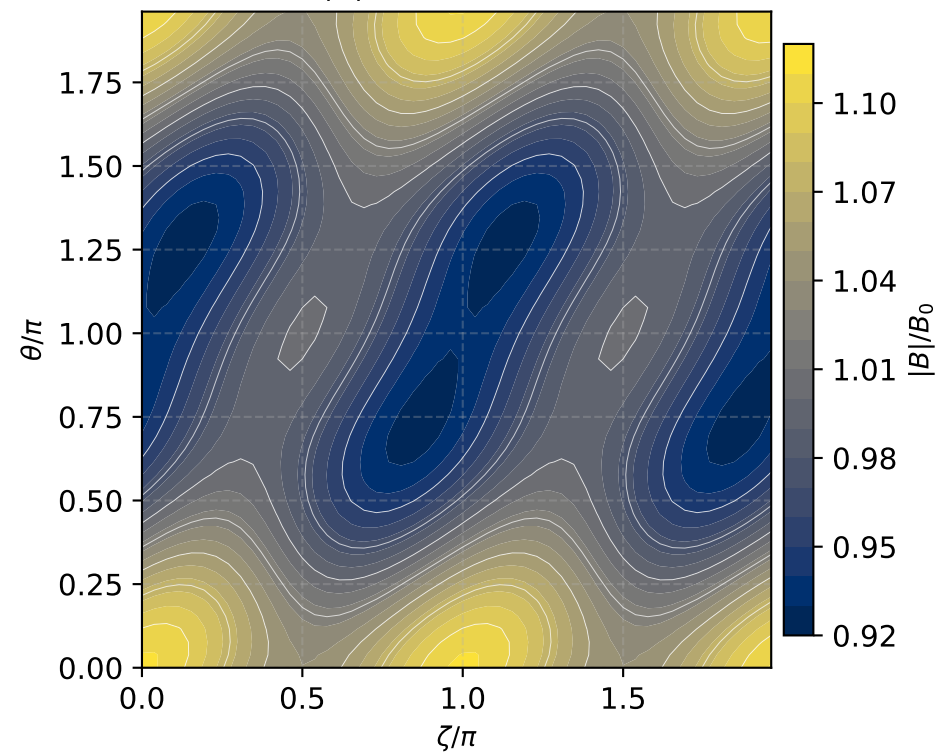
Gate summary
 Aspect target: 6.0
 Minimum iota: 0.41
 Operating iota floor: 0.70
 AD/FD gates passed: True
 Constraint gates passed: True
 Long-window gates passed: True
 Fixed-gradient Q reduction: 10.7%
 Trace tmax: 400

Final values
 QA constraints
 A=5.981, iota=0.749
 helical h=0.162
 QA residual=2.22e-04
 $\langle Q_i \rangle = 1.606e-02$
 QA + reduced NL Q
 A=5.972, iota=0.759
 helical h=0.163
 QA residual=2.36e-04
 $\langle Q_i \rangle = 1.434e-02$

LCFS $|B|$: QA constraints



LCFS $|B|$: QA + reduced NL Q



Scope and method

- Reduced max-mode-1 QA controls
- Aspect-ratio and two iota-floor constraints
- Long differentiable RK2 nonlinear ITG envelope
- Scalar/residual/observable AD-FD gates
- Production nonlinear claims still need long-window GK audits

Equations used in this panel

$J = \|r_A, r_i, r_{QA}, r_{reg}, r_Q\|^2$

$dE/dt = 2\gamma E - \alpha E^2$

$Q_i = W_i E$